Language Change and Population Turnover Explain Differences In Syntactic Biases Across Age Cohorts

Verbs can appear in multiple syntactic structures, sometimes creating ambiguity. In "pet the frog [with the feather]," the prepositional phrase (PP) could attach to the verb and signify the instrument used for the action, or to the noun, to modify the object of the verb. Individual verbs are biased to appear with one of the two structures, and listeners rely on these biases to disambiguate meanings [1,2]. However, English corpus analyses indicate that verb biases may differ across time [3,4] and dialects [5]. Older listeners did not appear to have the same verb biases as young adults, and these differences across the lifespan were predicted by changes in usage patterns from diachronic corpora [3]. Yet little attention has been paid to the relationship between age differences and population-level language change. What is the relationship between the diachronic patterns of language use of a linguistic community and the synchronic patterns of language use of individuals from different age cohorts within the same community?

Methods: Here, we used an agent-based model of Bayesian language learners in a social network to explore the relationship between language change with population turnover and syntactic biases. The model simulates the spread and usage of a syntactic construction throughout a population, where the construction is modeled as a binary feature (1 = instrument, 0 = modifier PP-attachment). Agents use a Beta distribution to track their beliefs about the usage distribution in a population (i.e., the likelihood of using an instrument construction) [6,7]. The Beta prior is defined by two parameters (α, β), and the *maximum a posteriori* (MAP) value of this distribution represents the syntactic biases of the agent. At every timestep, agents synchronously produce an utterance (1 or 0), listen and update their prior, and age. Speaking consists of a Bernoulli draw with the success probability (θ) of their MAP value (i.e., the agent's syntactic bias). When agents listen, they sample one utterance from a neighbor and update their prior via Beta-Binomial conjugacy ($\alpha_t = \alpha_{t-1} + k$, $\beta_t = \beta_{t-1} + n - k$, where k is the number of "1" utterances and n is the total number of utterances). We tested the influence of population size and expected lifespan (λ) across various starting conditions. Each agent had an 1/λ chance of dying every timestep, at which point it was replaced by a new agent with uninformative priors.

Results: Our model successfully replicated empirical patterns of verb biases, where the simulated linguistic populations developed a bias for a specific syntactic construction, and the bias at stabilization differed across populations (i.e., verbs), despite the fact that agents in all populations started with the same uninformative priors (Fig. 1). Smaller populations changed more quickly and had more extreme biases (further from 0.5) relative to larger populations. Short lifespans delayed, but did not prevent, stabilization of biases (defined as $\Delta bias < 0.005$ for at least 50 consecutive steps [7]; Fig. 2). In order to test for age-related differences in syntactic biases, we calculated the average correlation of the population-specific biases between independent age-cohort-specific samples across different expected lifespan (λ) values. Biases were more consistent in the older cohorts (Fig. 3). Biases in the younger cohort became more consistent as λ increased. The shortest lifespan (λ = 250) also led to the largest age-related differences in biases (Fig. 4). Each cohort's biases were the least similar to the younger cohort's biases and most similar to the older cohort's biases.

Conclusion: Our model demonstrates how different syntactic biases may emerge across populations/verbs, and critically, that population turnover is a sufficient condition for the emergence of age differences in syntactic biases. This work reveals the relationship between the timescales of language change and population turnover and their effects on synchronic language use by speakers of different ages within the same language community: if the rate of linguistic change of any linguistic feature is slow relative to the rate of population turnover, then usage should differ across age cohorts and, if it is fast, usage should be consistent across age cohorts.

References: [1] Snedeker & Trueswell, 2004, [2] Ryskin et al., 2017, [3] Cain & Ryskin, 2024, [4] Wolk et al., 2013, [5] Bresnan & Ford, 2010, [6] Griffiths & Kalish (2007), [7] Josserand et al., 2021

Figure 1: Population-level biases emerge at all population sizes. All agents start with uninformative priors.

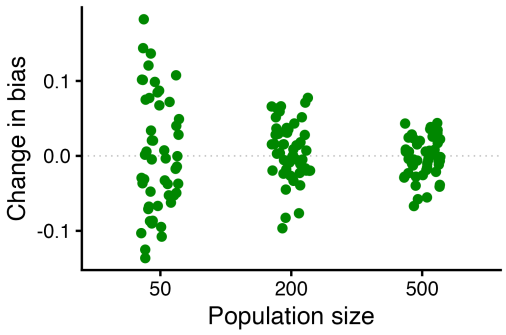

Figure 2: Time-to-stabilization decreases as the expected lifespan increases.

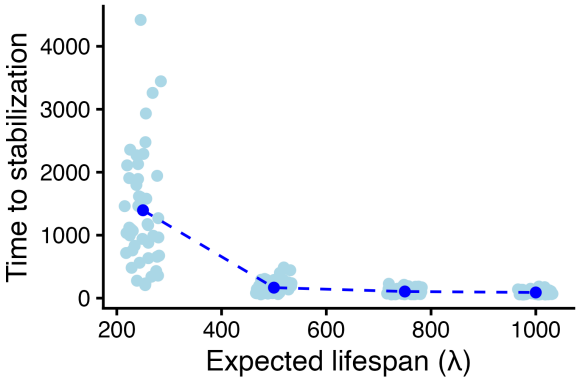

Figure 3: Correlation of biases between independent age-cohort-specific samples from the same age cohorts (200 repeated samples).

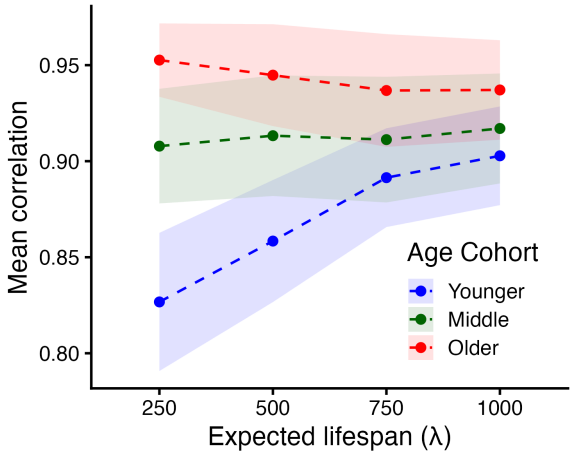
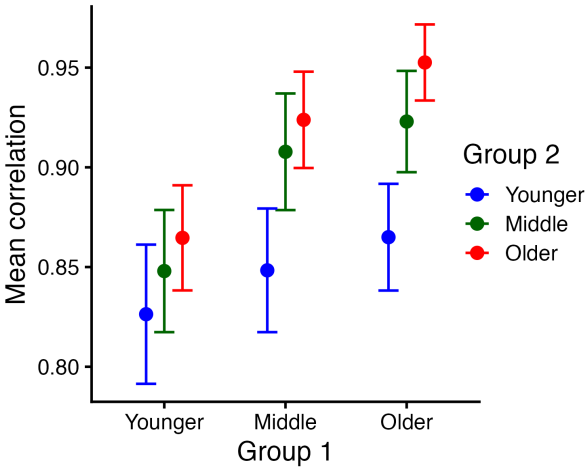
Figure 4: Correlation of biases between independent age-cohort-specific samples across age cohorts (200 repeated samples; λ = 250).